

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 1.4: Cell Division — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 1.4: Cell Division
Driving Question	DRIVING QUESTION: How does the body produce the enormous number of new cells needed for growth?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 From Newborn to Teenager: Where Do All the New Cells Come From?	Students compared a newborn's size with a 15-year-old's (e.g. a baby in Kisumu and a Form 1 learner) and noted that the teenager is far larger. Students also noticed that cells are too small to see, so a bigger body must contain many more cells, not just bigger ones. Teacher note: record student ideas on the Driving Question Board (DQB) without correcting them yet. Use WAIT TIME of 5 seconds after each question and cold-call 3 students.	Growth involves a huge increase in the NUMBER of cells. Eating ugali, sukuma wiki and beans supplies materials and energy but does not by itself explain where the new cells come from. At this point students do not yet know how new cells form, and this gap should be treated as a productive question. Teacher note: accept all predictions, such as 'cells split', 'food turns into cells' or 'cells just get bigger', and return to them in later lessons.	Students explained that the body must have a way of making many new cells from existing ones, but they could not yet say how. Their initial models should show a baby, a teenager and a question mark for the missing process. Teacher note: look for the misconception that food directly becomes cells, and keep a record of it to revisit in Lesson 10. Link back to the phenomenon by asking 'What must happen to the cell count between birth and age 15?'
Lesson 2 What Is Cell Division? Meaning of Cell Division and the Structure of the Chromosome	Students studied diagrams and models of a parent cell dividing into two daughter cells, and of a chromosome with two chromatids joined at a centromere. They noticed that the nucleus holds the chromosomes before division. Teacher note: a pipe-cleaner or string model of a chromosome works well when no projector is available.	Cell division is the process by which a parent cell divides to form new daughter cells, and this is how the body makes the many new cells needed for growth. A chromosome is made of DNA coiled with proteins, and a duplicated chromosome consists of two identical sister chromatids joined at the centromere. Genes are sections of DNA on chromosomes. Teacher note: stress that chromatids are identical copies, since students often confuse them with homologous chromosomes.	Students used this knowledge to propose that growth happens as existing cells divide, so one cell becomes two, then four, then eight. DNA must be copied before division so each daughter cell gets a full set of instructions. Teacher note: ask 'Why must DNA be copied first?' and wait 5 seconds. Add new questions to the DQB, such as 'How exactly do the chromosomes get shared equally?'
Lesson 3 Types of Cell Division:	Students examined diagrams, photographs or models of cells in each stage of mitosis	Mitosis has four stages, prophase, metaphase, anaphase and telophase,	Students explained that mitosis answers part of the driving question, since one cell

Mitosis and Its Stages	and arranged them in order. They noticed chromosomes condensing, lining up in the middle, separating and moving to opposite poles. Teacher note: a card-sort activity in groups of four works well, and learners should justify their order using evidence from the images.	followed by cytokinesis. It produces two genetically identical daughter cells, each with the same number of chromosomes as the parent. In prophase chromosomes condense, in metaphase they line up at the equator, in anaphase chromatids separate, and in telophase two nuclei form. Teacher note: a mnemonic such as PMAT helps, but check that students understand what happens and do not just memorise the letters.	becomes two identical cells and repeated rounds produce billions of cells. The equal sharing of chromatids ensures each new cell has a complete set of DNA. Teacher note: probe with 'What would happen if chromatids did not separate evenly?' Update the class model with a mitosis cycle arrow.
Lesson 4 Meiosis and Its Stages: Why Do Sperm and Egg Cells Have 23 Chromosomes?	Students compared the human body cell chromosome number (46) with sperm and egg cells (23) using diagrams and chromosome models. They followed the two divisions, meiosis I and meiosis II, and counted the cells produced. Teacher note: use different coloured pipe cleaners for maternal and paternal chromosomes to show pairing and separation.	Meiosis is a division in reproductive organs that has two divisions (meiosis I and meiosis II) and produces four cells, each with half the chromosome number of the parent (haploid). In meiosis I homologous chromosomes separate, and in meiosis II chromatids separate. Fertilisation restores 46 chromosomes. Teacher note: stress that meiosis makes gametes, not growth cells, and that the daughter cells are genetically different.	Students explained that meiosis is not the process that makes the large number of cells for growth, because it halves the chromosome number and occurs only in reproductive organs. It keeps the chromosome number constant across generations, because $23 + 23 = 46$ at fertilisation. Teacher note: ask students to contrast mitosis and meiosis in a two-column table. Revisit the DQB to separate growth questions from reproduction questions.
Lesson 5 Significance of Mitosis and Meiosis: Case Studies from Growth, Healing, Roots and Offspring	Students read case studies on a child's growth, a wound healing on the skin, roots growing in a maize seedling or sweet potato cutting, and a child resembling but not identical to the parents. They identified the type of division involved in each case. Teacher note: assign one case to each group and use a jigsaw share-out.	Mitosis produces genetically identical diploid cells for growth, repair and asexual reproduction in plants, which is how a child makes billions of new cells. Meiosis produces gametes with variation, so offspring differ from their parents and from each other. Teacher note: link variation to the idea that siblings in the same Kenyan family look similar but not identical.	Students used evidence from the cases to explain that growth and healing depend on mitosis, since new cells must match the old ones. They argued that meiosis explains variation and not growth. Teacher note: use a claim-evidence-reasoning frame, and cold-call 3 groups to share. Update the model with labels showing 'mitosis: growth and repair' and 'meiosis: gametes'.
Lesson 6 Budding and Binary Fission: Making New Cells and Organisms from One Parent	Students observed images or a demonstration of yeast budding (e.g. yeast used in making mandazi or brewing) and bacteria dividing by binary fission. They noticed that the buds are smaller than the parent yeast cell, while the two bacterial cells are equal. Teacher note: a simple yeast and sugar mixture in warm water shows yeast activity if a microscope is available.	Budding gives unequal parent and daughter cells (yeast), while binary fission gives two equal cells (bacteria). Both involve copying genetic material before dividing, and both are forms of asexual reproduction in which one parent produces genetically identical offspring. Teacher note: clarify that in these organisms cell division makes new organisms, while in humans it makes more cells in one body.	Students explained that copying DNA and then dividing is a common pattern across living things, supporting the idea that new cells arise from existing cells. They compared these methods with mitosis in a human body. Teacher note: ask 'How is binary fission similar to mitosis, and how is it different?' and give 5 seconds of wait time. Add to the DQB: 'Do all cells divide in the same way?'
Lesson 7 Microscope Practical on	Students prepared or viewed slides of onion root tips under the microscope, and counted	Mitosis is how existing cells divide to form two identical daughter cells, each with a full	Students used their counts as evidence that growing regions contain more dividing cells,

Mitosis: Catching Growth in Action in Onion Root Tips	cells in interphase and in the different stages of mitosis. They noticed many cells dividing near the tip and few further back. Teacher note: if no microscope or stained slide is available, use prepared photographs. Remind students to use low power first, then high power, and to draw only what they see.	set of chromosomes. Growing tissues such as root tips have many cells in division. Most cells in a sample are in interphase because it is the longest phase. Teacher note: a tally of cells in each stage can be used to show that dividing is a short but busy part of the cell cycle.	so cell division is directly linked to growth. They related the root tip of an onion or maize to growth in children. Teacher note: ask 'What evidence from your slide shows that growth is happening here?' Students should add the practical results to their model as real data.
Lesson 8 Why Cell Division Matters: From Baby to Teenager, and Bananas from a Test Tube	Students studied examples of how cell division works in practice, such as growth from baby to teenager, wound repair and tissue culture of bananas in Kenyan laboratories (e.g. banana tissue culture for farmers in Kisii, Meru and the Rift Valley). They noted that new plants grow from a few cells. Teacher note: bring in a picture of tissue-cultured banana plantlets to make the example concrete.	Cell division by mitosis produces genetically identical cells for growth, repair and asexual reproduction, while meiosis produces gametes and variation. Tissue culture uses mitosis to make many identical, disease-free banana plants from small pieces of tissue. Teacher note: connect this to food security and farmer incomes, which are relevant to many Kenyan learners.	Students explained why tissue culture banana plants are identical to the parent plant, since they come from mitosis. They also explained that the same process makes the new cells in a growing child. Teacher note: ask students to give a claim and evidence in pairs, then cold-call 3 pairs. Update the DQB by marking the questions that can now be answered.
Lesson 9 Consolidation and Digital Animations: Comparing Mitosis and Meiosis to Explain Growth	Students watched animations, or followed teacher-drawn step-by-step diagrams where no devices are available, of mitosis and meiosis side by side. They noticed that mitosis has one division and gives two cells, while meiosis has two divisions and gives four cells. Teacher note: pause the animation after each stage and ask students to predict the next step before continuing.	Mitosis produces genetically identical diploid cells for growth and repair, such as bone, skin and the root tips of maize. Meiosis produces haploid gametes that differ genetically, and these combine at fertilisation. Teacher note: use a comparison table with the headings where it occurs, number of divisions, number of cells, chromosome number, genetic identity and purpose.	Students used the comparison to explain that only mitosis can account for the huge increase in cell number needed for growth. Meiosis is needed for reproduction, not growth. Teacher note: listen for the common error of saying meiosis makes skin cells, and correct it with evidence. Students should revise the class model to show both processes in their correct places.
Lesson 10 Final Model and Return to the Phenomenon: How Does the Body Make the Billions of New Cells Needed for Growth?	Students reviewed their DQB, earlier models, practical results and case studies, and compared their first ideas from Lesson 1 with what they now know. They noticed how much their understanding had changed. Teacher note: display the Lesson 1 predictions next to the final models to make the change visible.	Growth happens mainly because the number of cells increases. Existing cells copy their DNA and divide by mitosis into two identical daughter cells, and repeating this many times produces billions of cells. Food provides the energy and materials for this process but does not turn directly into cells. Meiosis makes gametes and is not responsible for growth. Teacher note: emphasise that cells also enlarge and specialise, but cell division supplies the increase in number.	Students gave a complete explanation of the phenomenon: a newborn grows into a teenager as cells repeatedly copy their DNA and divide by mitosis, using nutrients from foods such as ugali, beans and sukuma wiki. They used evidence from the onion root tip practical, the case studies and the tissue culture bananas. Teacher note: assess using a labelled final model and a written or oral explanation, and give a 5-second WAIT TIME after each question. Check that students have corrected earlier misconceptions about food becoming cells.

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.
